

Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

- The NYC taxi trip data was loaded from multiple parquet files and combined into a single dataframe for analysis. A sample of the data was used initially to reduce memory usage and improve processing speed during exploratory analysis. Took sample 0.05 of hour_data,
- After loading the files, the datasets were concatenated together to create a unified dataset containing trip-level taxi information such as pickup/dropoff locations, fare amount, trip distance, passenger count, payment type, and timestamps.

```
df = pd.read_parquet('../Databases and  
Dictionary/trip_records/2023-1.parquet')  
df.info()
```

Index: 3041714 entries, 0 to 3066765

Data columns (total 19 columns):

#	Column	Dtype
0	VendorID	int64
1	tpep_pickup_datetime	datetime64[us]
2	tpep_dropoff_datetime	datetime64[us]
3	passenger_count	float64
4	trip_distance	float64
5	RatecodeID	float64
6	store_and_fwd_flag	object
7	PULocationID	int64
8	DOLocationID	int64
9	payment_type	int64
10	fare_amount	float64
11	extra	float64
12	mta_tax	float64
13	tip_amount	float64
14	tolls_amount	float64
15	improvement_surcharge	float64
16	total_amount	float64
17	congestion_surcharge	float64
18	airport_fee	float64

dtypes: datetime64[us](2), float64(12), int64(4), object(1)
memory usage: 464.1+ MB

2. Data Cleaning

Fixing Columns

- To ensure data integrity and improve the quality of analysis, a comprehensive data cleaning process was performed on the sampled dataset.
- **Fix the index**

The dataset index was reset to ensure sequential indexing after combining the files and removing invalid rows. This helped maintain consistency during filtering and analysis operations.

Below unnecessary columns were dropped:

store_and_fwd_flag: Not useful for analysis; indicates if the trip record was held in vehicle memory before sending to the server.

mta_tax: Fixed tax amount, does not provide variability.

tolls_amount: Not relevant for most trips, as tolls are not always applicable.

Fix data type of column

From the table above, we can observe that a few columns do not have the correct data types and need to be fixed:

- **RatecodeID** is currently stored as `float64`, but according to the data dictionary it contains categorical values such as 1, 2, 3, 4, 5, and 6. Therefore, it should be converted to an integer type.

passenger_count is also stored as `float64`, but the number of passengers cannot be in decimal form, so it should be converted to `int64`.

- **Combine the two airport_fee columns**

Columns Merging

There are two airport fee columns: `airport_fee` and `Airport_fee`. This is possibly an error in naming columns. These two columns were merged by adding the values of both columns and the new column was named as `airport_fee_combined`.

Fixing Negative Values

During the data cleaning process, it was observed that certain financial columns contained negative values, which were not expected in the dataset.

	Column	Negative-Value(%)
0	VendorID	0.000
1	tpep_pickup_datetime	0.000
2	tpep_dropoff_datetime	0.000
3	passenger_count	0.000
4	trip_distance	0.000
5	RatecodeID	0.000
6	PULocationID	0.000
7	DOLocationID	0.000
8	payment_type	0.000
9	fare_amount	0.000
10	extra	0.000
11	tip_amount	0.000
12	improvement_surcharge	0.003
13	total_amount	0.003
14	congestion_surcharge	0.001
15	airport_fee_combined	0.001

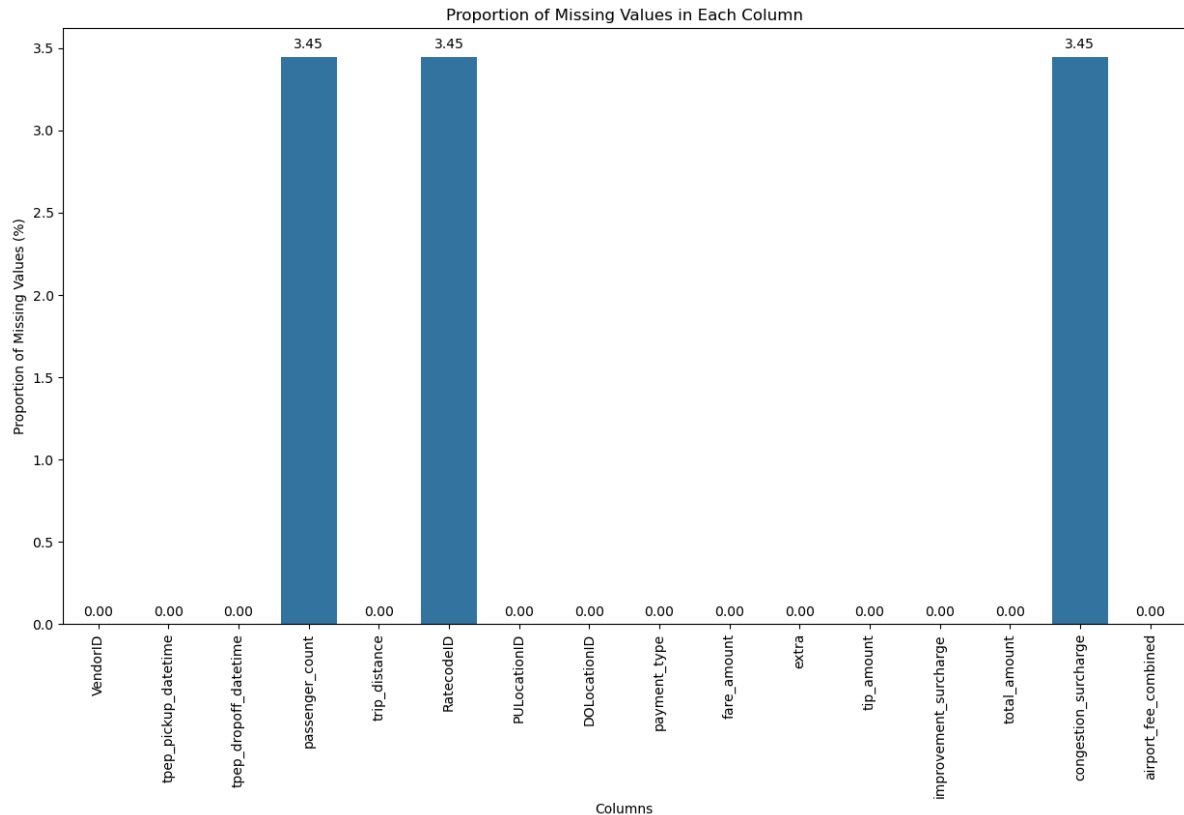
- Let's fix the columns with negative values:
improvement_surcharge
total_amount
congestion_surcharge
Airport_fee_combined
- We will replace the negative values with the absolute value of that column.
- After replacing negative value with absolute, here the negative percentage, i.e 0

	Column	Negative-Value(%)
0	VendorID	0.0
1	tpep_pickup_datetime	0.0
2	tpep_dropoff_datetime	0.0
3	passenger_count	0.0
4	trip_distance	0.0
5	RatecodeID	0.0
6	PULocationID	0.0
7	DOLocationID	0.0
8	payment_type	0.0
9	fare_amount	0.0
10	extra	0.0
11	tip_amount	0.0
12	improvement_surcharge	0.0
13	total_amount	0.0
14	congestion_surcharge	0.0
15	airport_fee_combined	0.0

- Handling Missing Values

Missing values were analysed for each column to understand the quality of the dataset. Columns with very low missing percentages were handled using imputation techniques, while maintaining the integrity of the original data distribution.

Find the proportion of missing values in each column

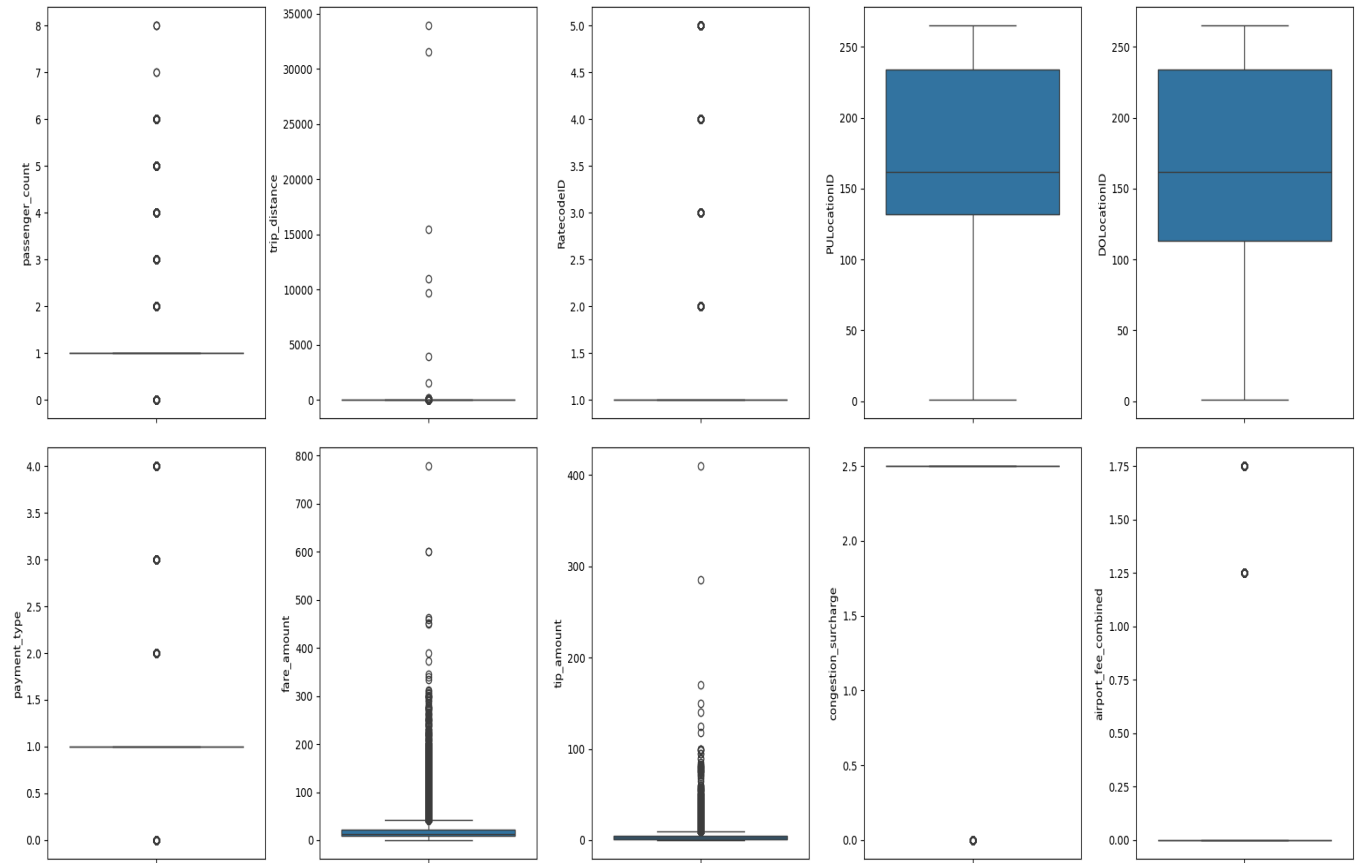


- **Handling missing values in passenger_count**
 - Missing values in passenger_count were imputed using the mode/median value since passenger counts are discrete numerical values and most trips involved a small number of passengers.
 - After handling missing value percentage is 0%
 - **Handle missing values in RatecodeID**
 - Missing values in RatecodeID were filled using the most frequent category because RatecodeID represents a categorical fare classification.
- There was ratcode id 99 which was not a category, so replace it with the mode first value i.e 1

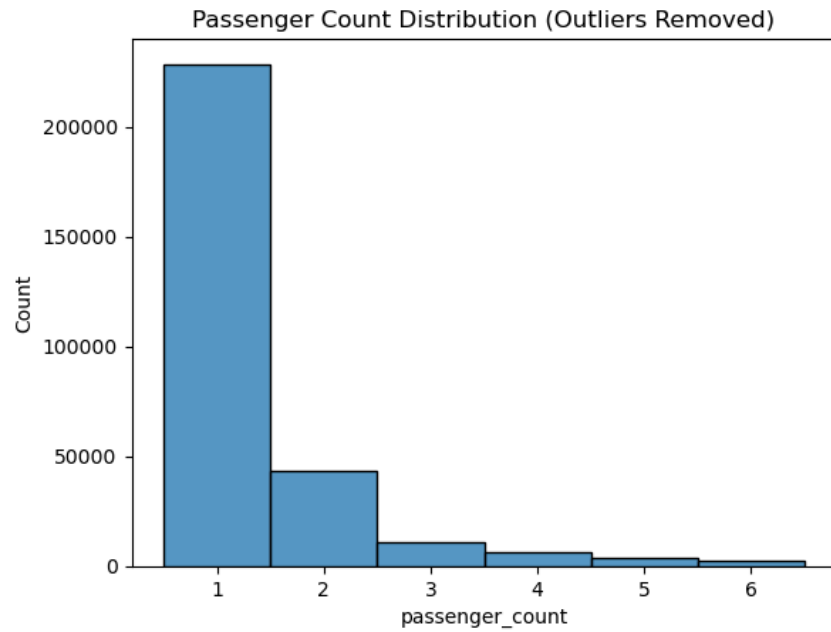
```
Missing values before: 10343
The most frequent RatecodeID is: 1
Missing values after: 0
```

```
RatecodeID
1      285388
2      11346
5       1718
3        932
4        616
Name: count, dtype: Int64
```

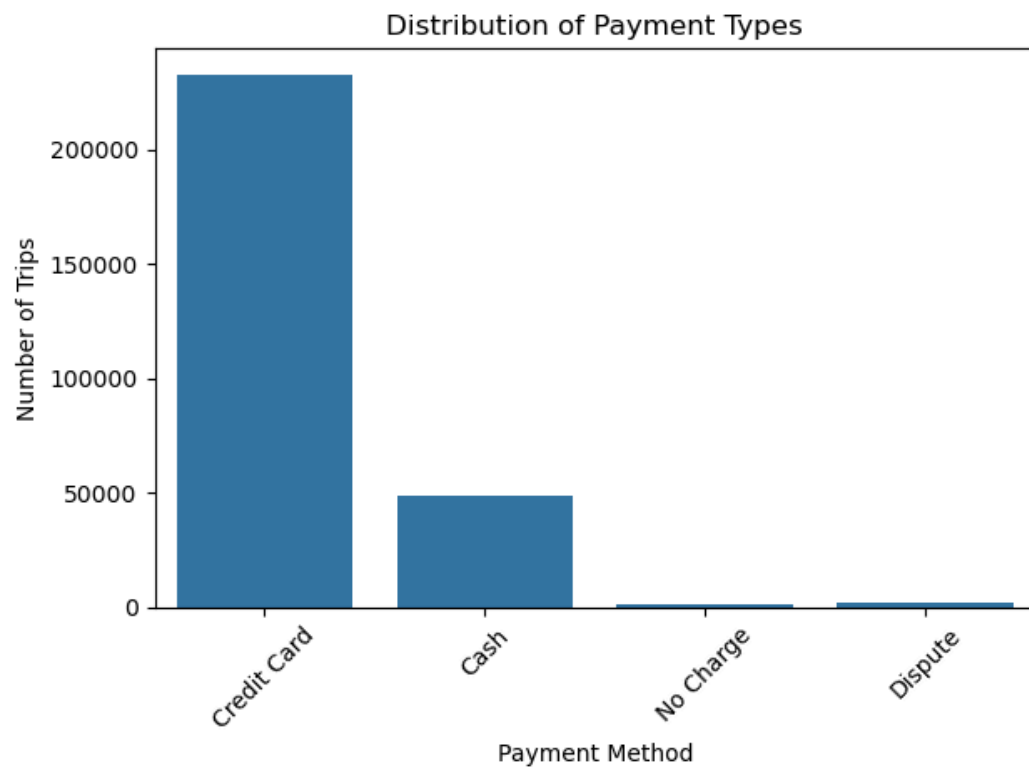
- **Impute NaN in congestion_surcharge**
Impute the missing values in 'congestion_surcharge' with the median value
- **Handling Outliers and Standardising Values**
- Outlier analysis was performed on columns such as trip_distance, fare_amount, payment_type, and tip_amount. Using boxplot checked for potential out of place values in various columns
- Several logically inconsistent records were identified and removed, including:
 - Trips with extremely large trip distances
 - Trips with zero distance and zero fare between different pickup and dropoff zones
 - Invalid payment types
 - Trips with unrealistic fare values for very short distances



- Remove `passenger_count > 6` and draw a histplot.



- Remove entries where payment_type is 0, and show after countplot



- Standardising column names by converting them all to lowercase

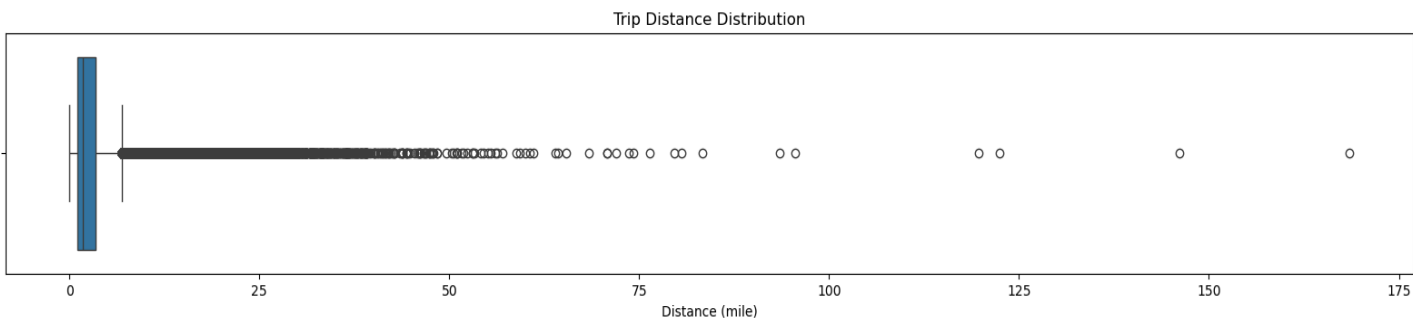
- Standardising data types for categorical ID columns
- Converting these from numbers to 'category' for ['vendorid', 'ratecodeid', 'pulocationid', 'dolocationid', 'payment_type']

```
Standardised Columns: ['vendorid', 'tpep_pickup_datetime', 'tpep_dropoff_datetime', 'passenger_count', 'trip_distance', 'ratecodeid', 'pulocationid', 'dolocationid', 'payment_type']

Data types after standardisation:
vendorid      category
ratecodeid    category
pulocationid  category
dolocationid  category
payment_type  category
dtype: object
```

- Found rows which are not from 2023 as we are only interested in 2023 data and removed other year data.

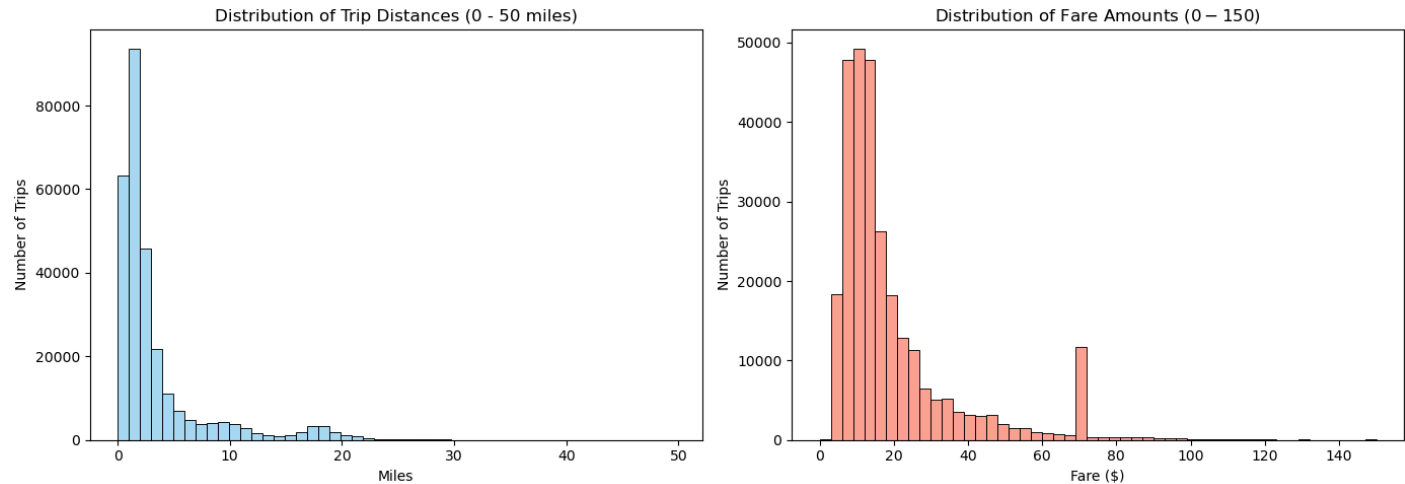
Handling Outlier trip_distance with boxplot



- Trips with extremely short distances combined with unusually high fares were treated as suspicious records and removed from the dataset.
- Assuming that taxis that travel more than (120 miles) are incorrect records as these are extreme outliers.
- Entries where `trip\_distance` and `fare\_amount` are 0 but the pickup and dropoff zones are different (both distance and fare should not be zero for different zones)

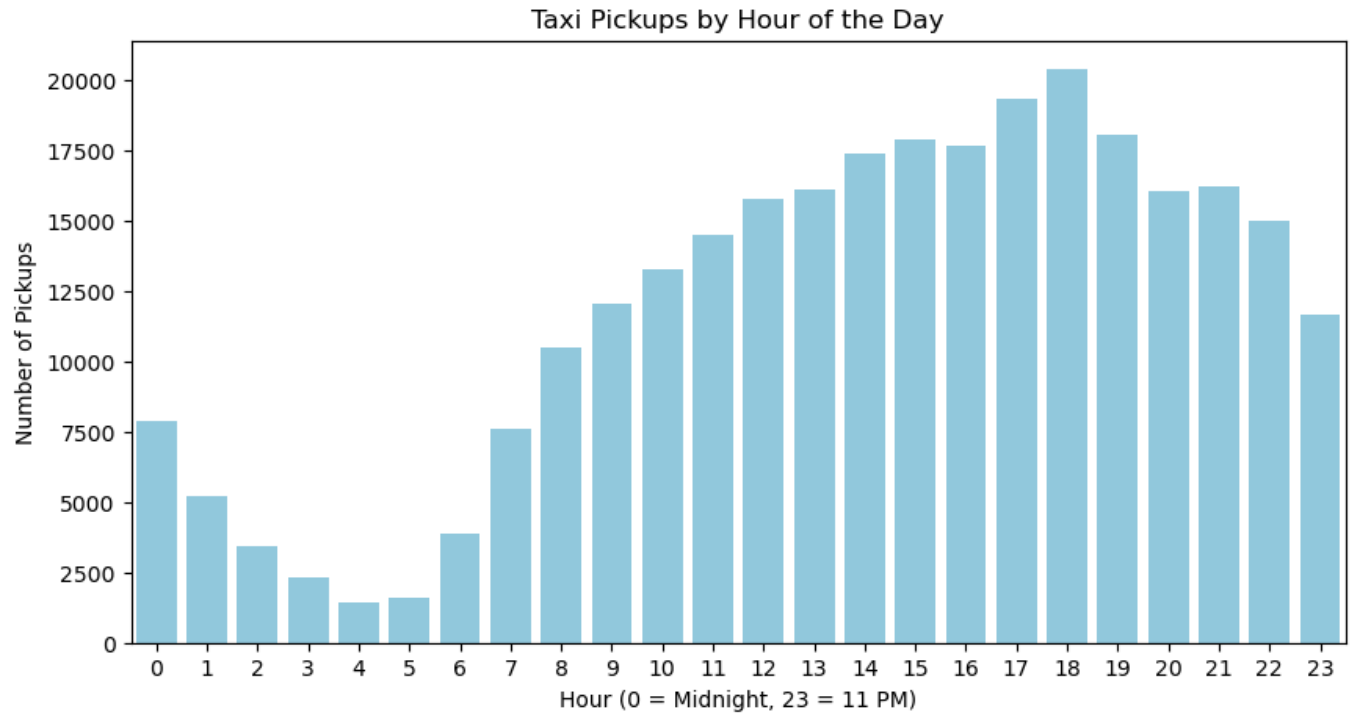
```
Invalid trips before removing : 40
Invalid trips after removing : 0
```

- Visualising after cleaning outlier by histplot

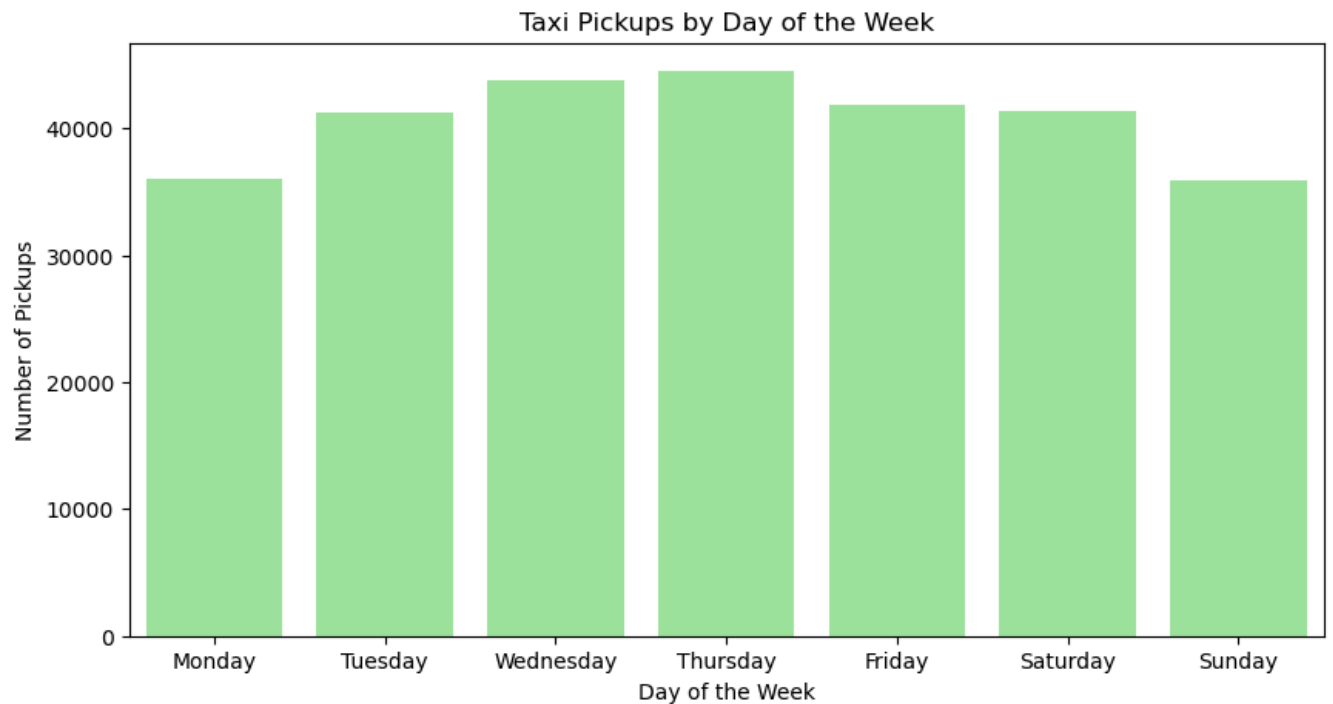


Exploratory Data Analysis

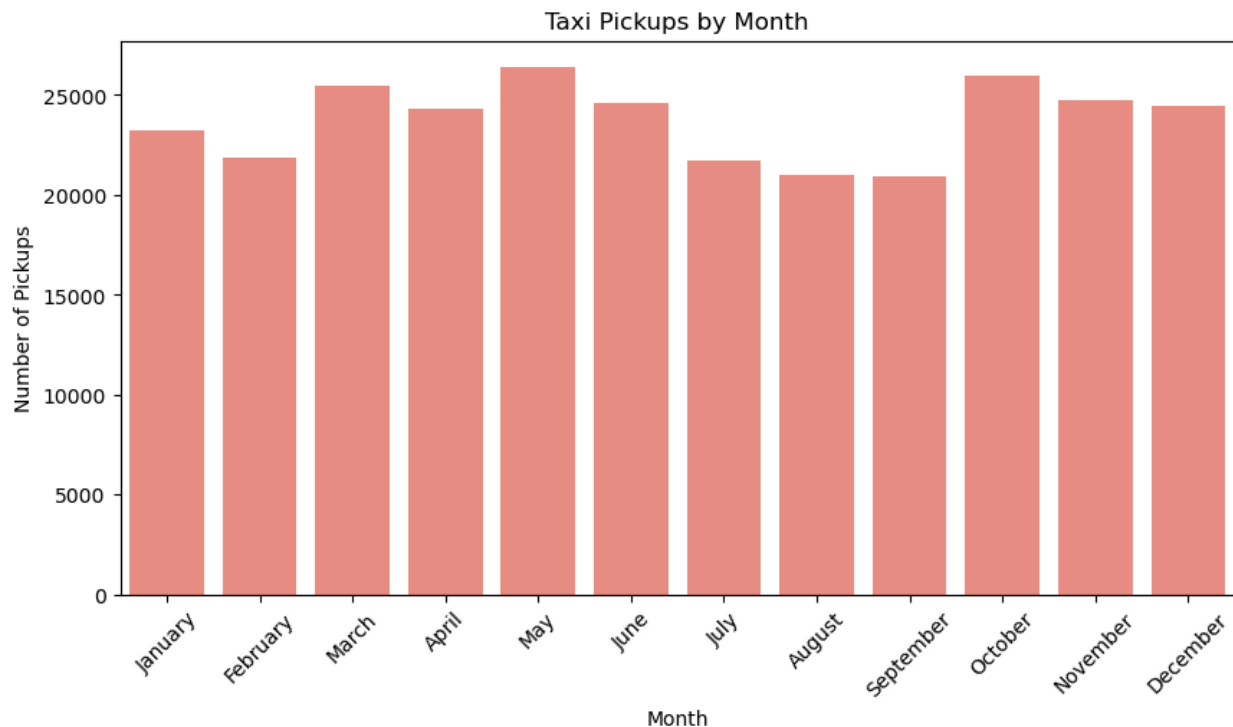
- General EDA: Finding Patterns and Trends
- **Classify variables into categorical and numerical**
Converted the column in categorical and numerical as part of data cleaning
- **Analyse the distribution of taxi pickups by hours, days of the week, and months**
- First created few column to use it further from "tpep_pickup_datetime" with new columns as [pickup_hour, pickup_day_num, pickup_month_num, pickup_day, pickup_month] extended by "_num" for numerical value
- Now showing hourly trend in taxi pick up by countplot using **pickup_hour** as x axis and data existing data frame



- Show the daily trends in taxi pickups (days of the week) using countplot



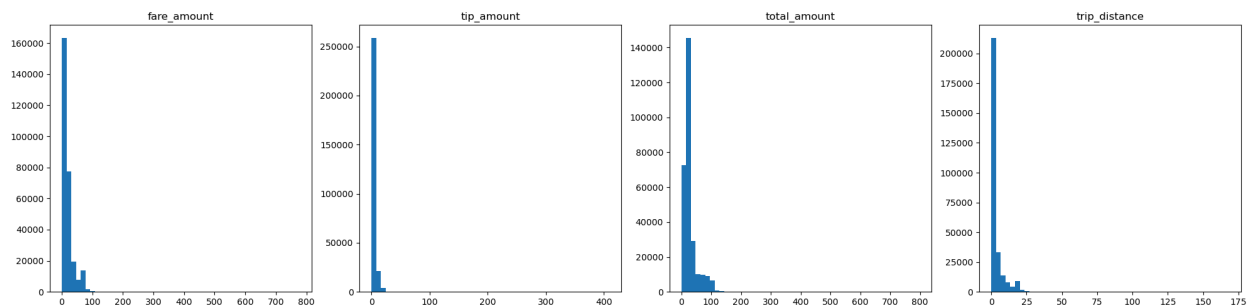
- Show the monthly trends in pickups



- Financial Analysis

Checking financial parameters like `fare\_amount`, `tip\_amount`, `total\_amount`, and also `trip\_distance` for zero or negative value

Plot for above parameter



- Do you think it is beneficial to create a copy DataFrame leaving out the zero values from these?

Yes, it is very beneficial.

- Zero values in columns like trip_distance and fare_amount are typically recording errors (e.g., the meter was engaged but the GPS failed to track movement, or the passenger cancelled immediately after getting in)
- If we include these zeros in our statistical analysis or machine learning models, they act as heavy anchors—artificially pulling down our averages and ruining our correlation calculations. Creating a copy of the DataFrame without these zeros is the

smartest approach because it allows us to analyze the "true" trips accurately, without permanently deleting the original records (which we might still want later just to count total taxi dispatches)

- Removed 0 and negative values

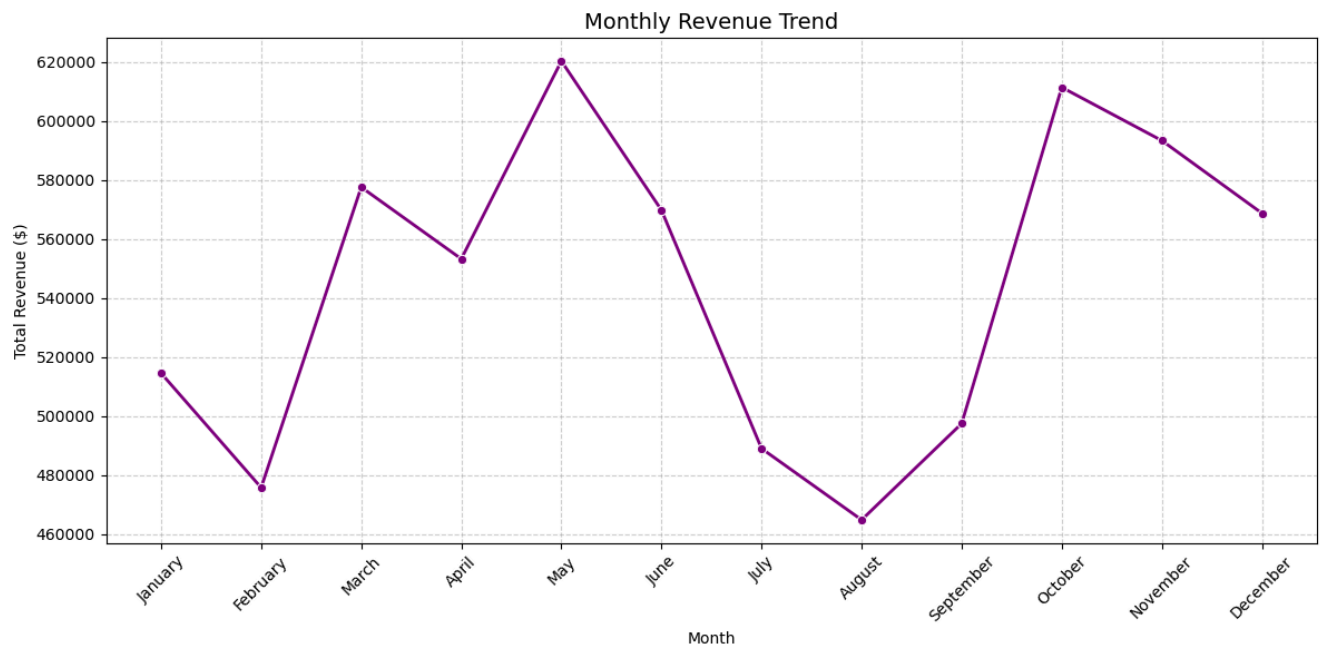
```
Original DataFrame size: 284,794 trips
Zero-free DataFrame size: 220,262 trips
Removed 64,532 problematic zero-value trips.
```

- **Filter out the zero/negative values in fares, distance and tip**

```
Total trips before filtering: 284,794
Total trips after filtering: 220,262
```

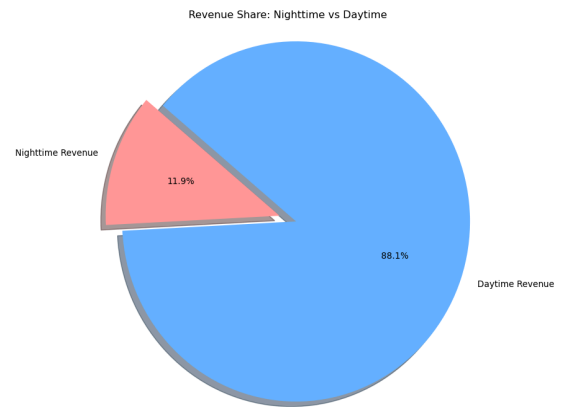
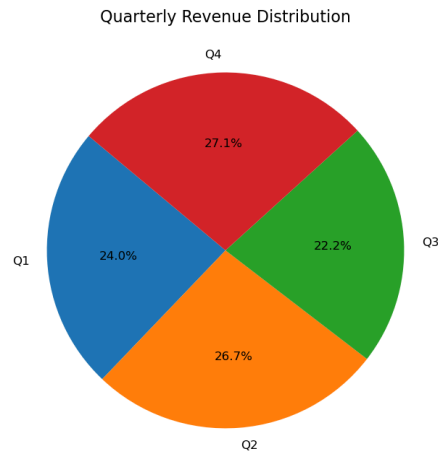
- **Analyse the monthly revenue trends**

Group data by month and analyse monthly revenue

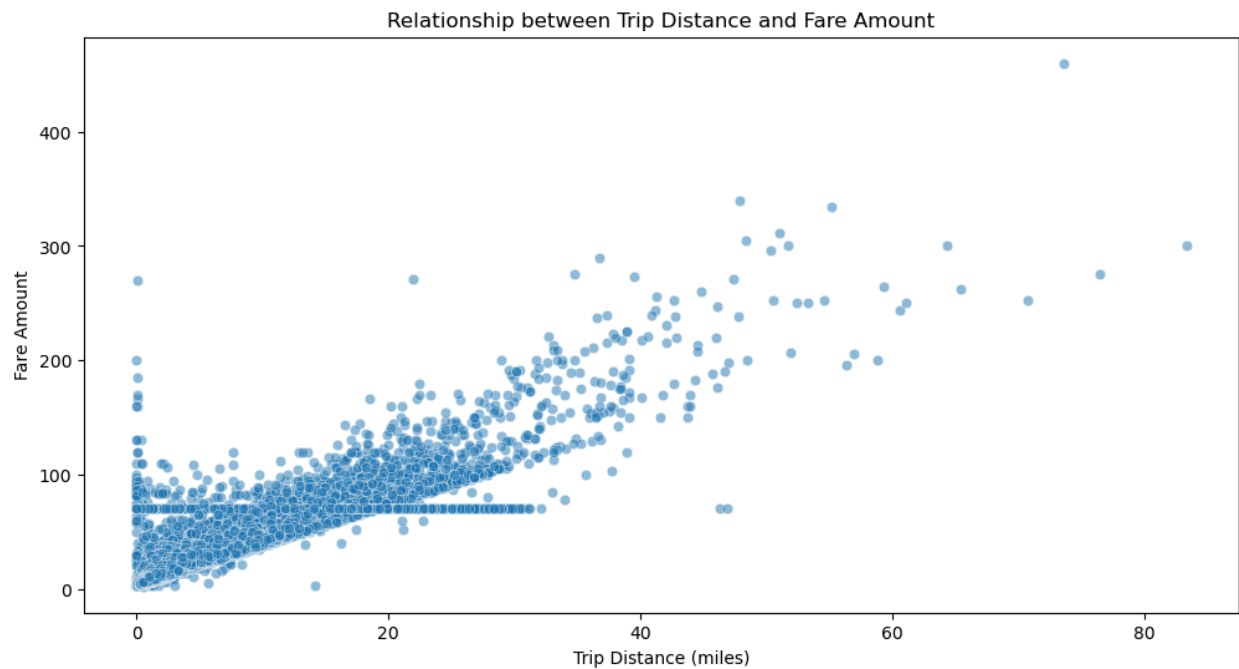


- **Find the proportion of each quarter's revenue in the yearly revenue**

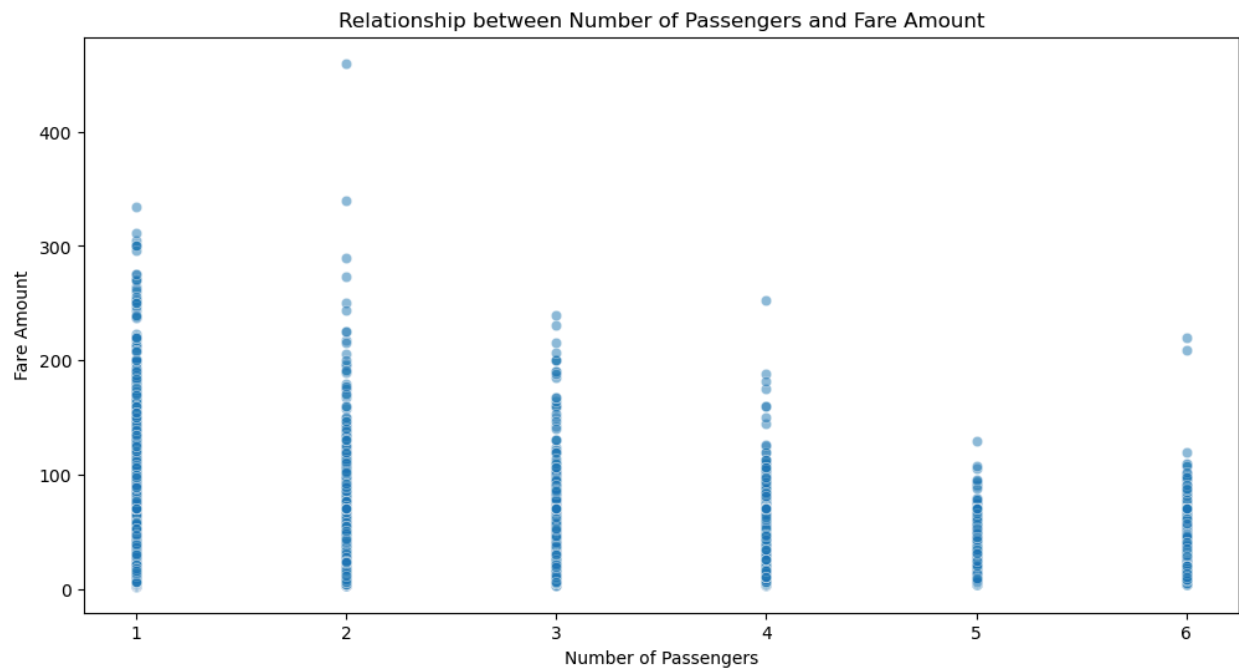
The key insights from the charts are that the highest revenue was generated in Quarter 4, with October being the peak month. Conversely, the lowest revenue occurred in Quarter 3, with August being the bottom month. Most of the revenue was generated by the trips completed in daytime.



- **Analyse and visualise the relationship between distance and fare amount**
Correlation between Trip Distance and Fare Amount: 0.95

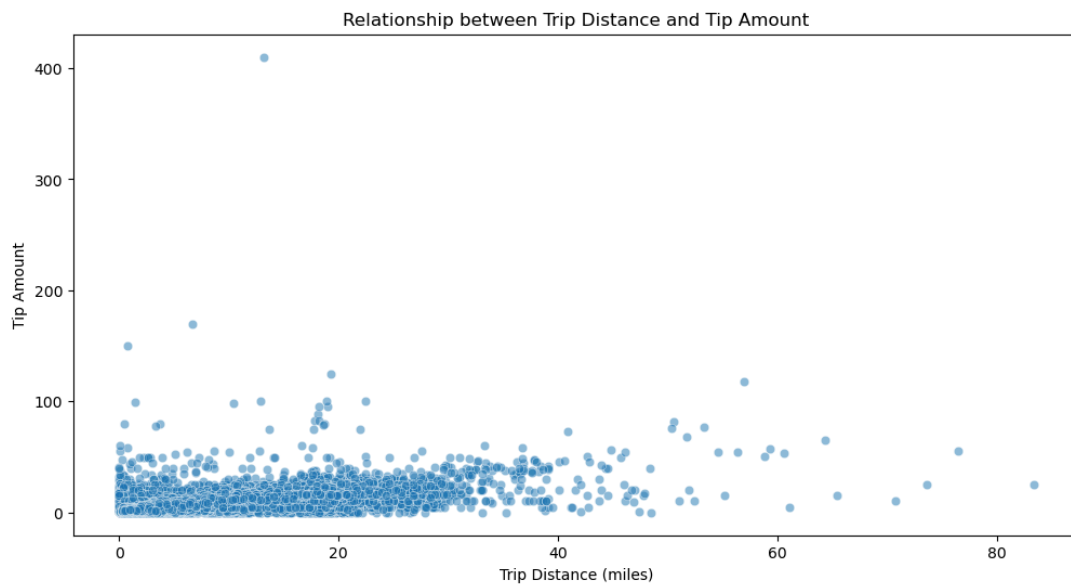


- **Relationship between fare and number of passengers**
Correlation between Number of Passengers and Fare Amount: 0.04

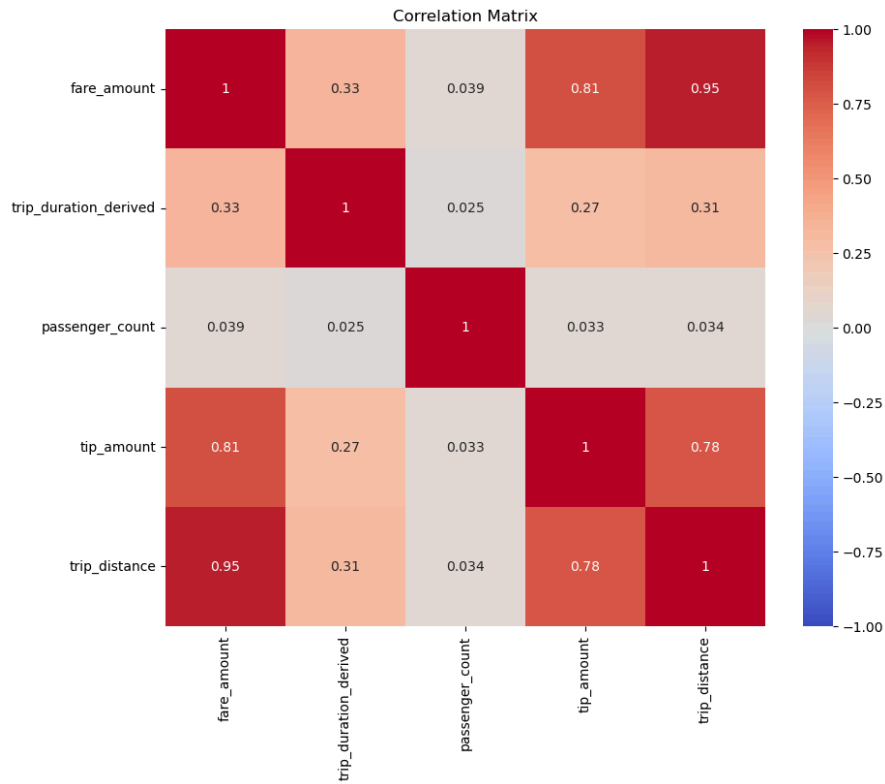


- **Relationship between tip and trip distance**

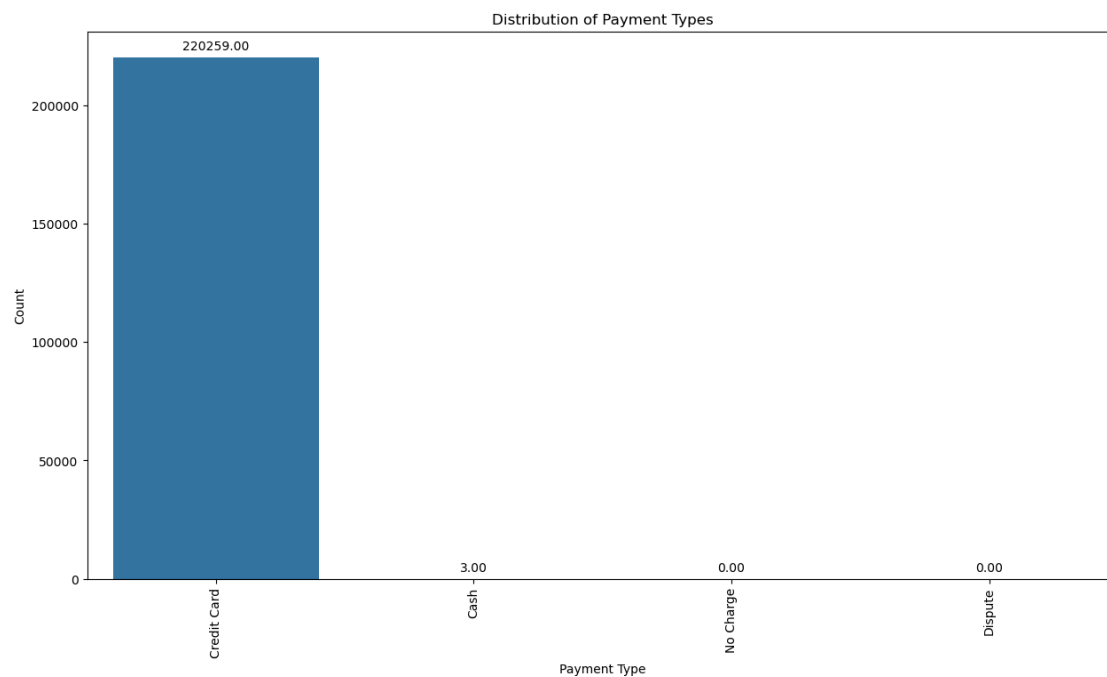
Correlation between Trip Distance and Tip Amount: 0.78



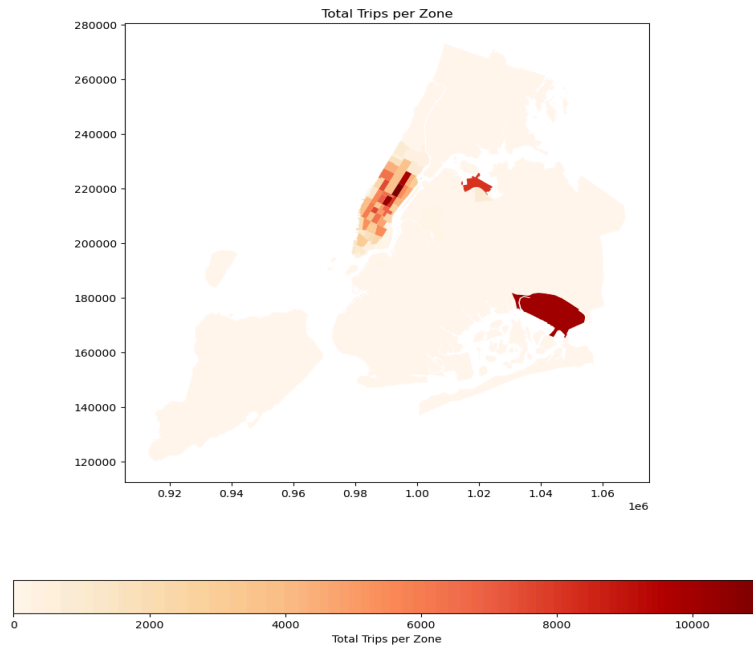
- **Correlation b/w fare_amount, trip_duration_derived, passenger_count, tip_amount, trip_distance**



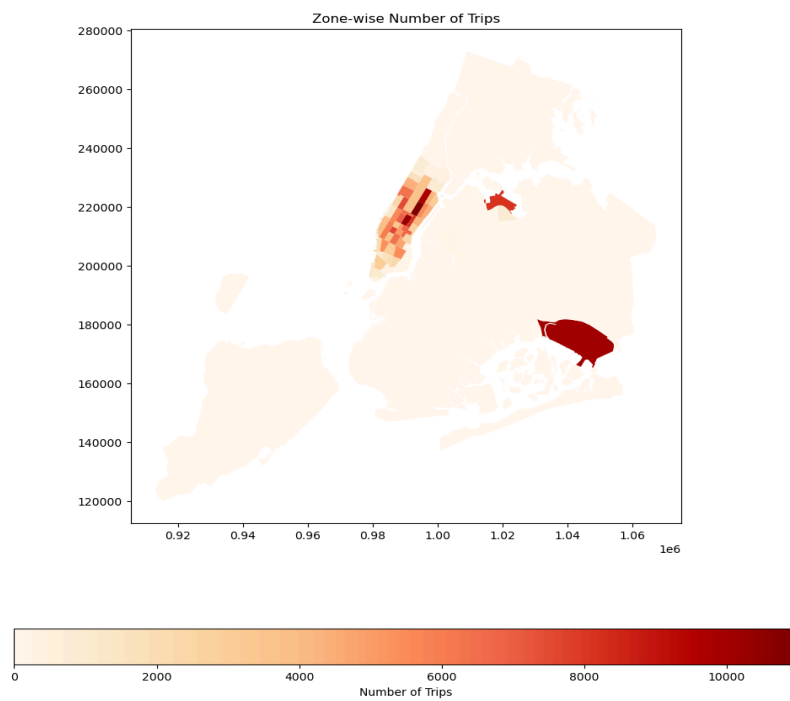
- **Analyse the distribution of different payment types**
- After seeing plot most of the payments were done by Credit card, and few with cash



- **Total Trip per zone**



- **Zone wise number of trips**



- **Top 5 Zone by trip count are following:**

	zone	trip_count
236	Upper East Side South	10999.0
160	Midtown Center	10381.0
131	JFK Airport	10017.0
235	Upper East Side North	9755.0
161	Midtown East	8165.0

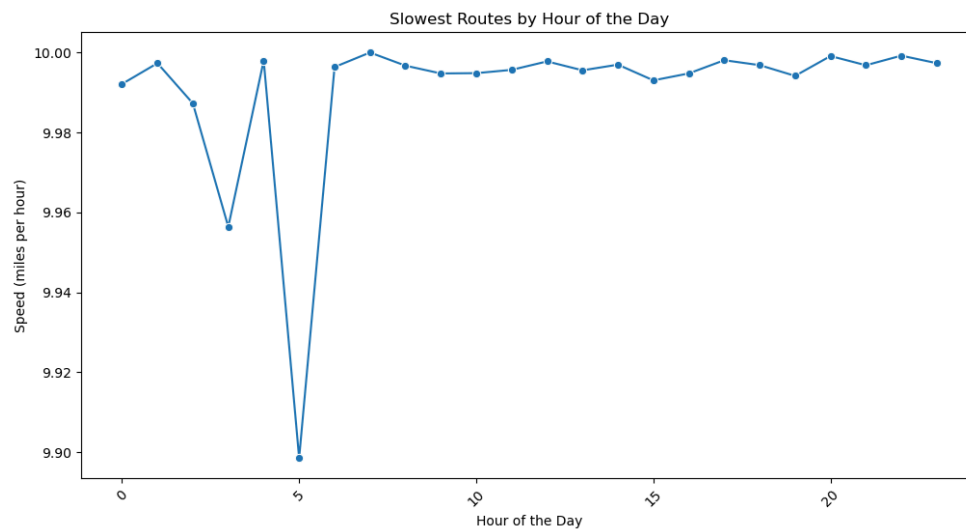
- Operational Efficiency

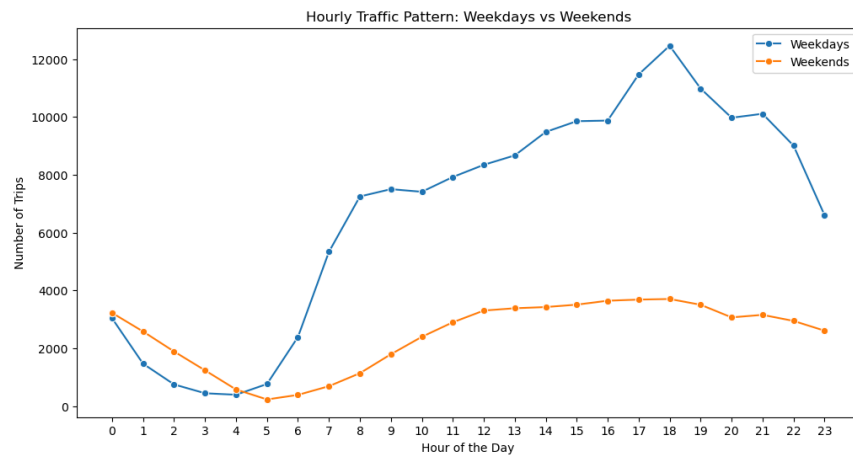
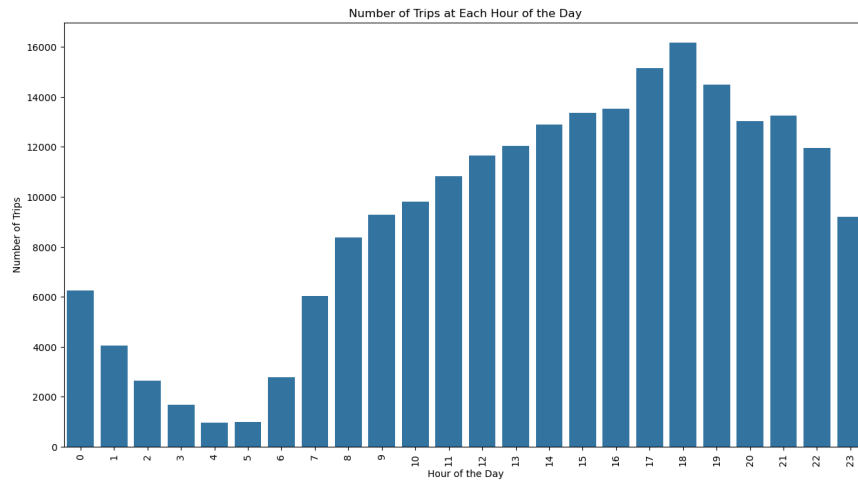
Traffic and trend analysis

- The busiest hour is 18:00 with 16166 trips.
- The speed remains relatively constant, fluctuating between approximately 9.988 and 10.000 miles per hour. The slowest routes do not vary significantly in speed throughout the day, indicating consistent traffic conditions on these routes.
- Peak hours with higher trip volumes are between morning and evening rush hours. Weekdays show higher trip volumes during commuting hours (e.g., 7-9 AM and 4-7 PM), while weekends have more consistent trip volumes throughout the day, peaking in the afternoon.

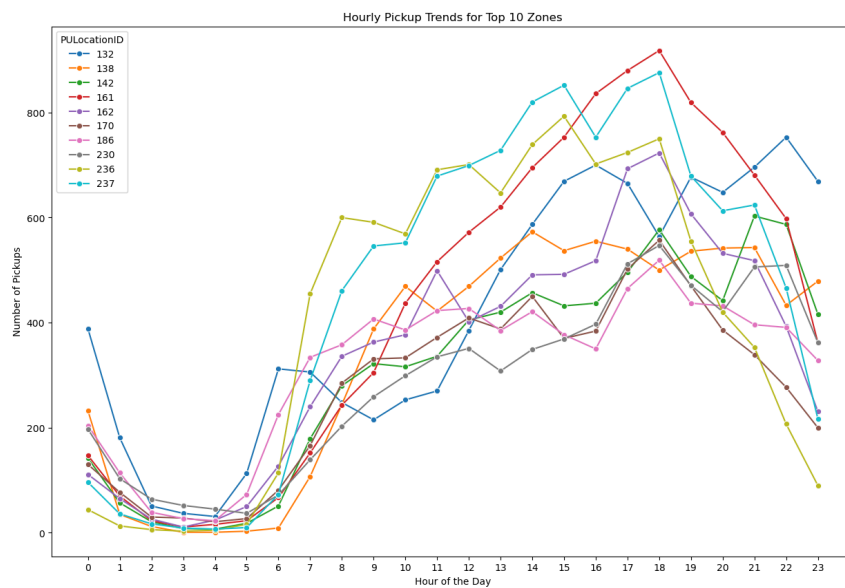
Actual number of trips in the five busiest hours:

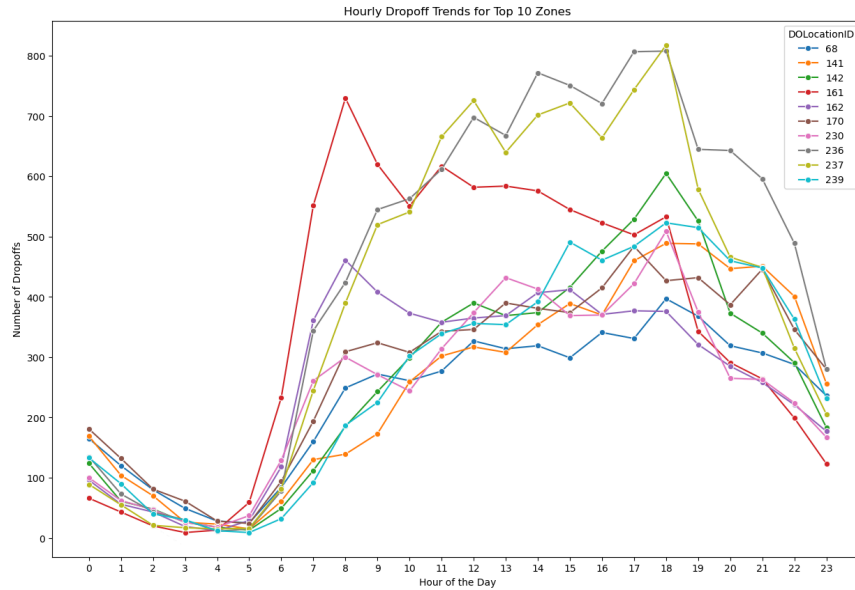
```
pickup_hour
18    16166
17    15152
19    14486
16    13518
15    13361
Name: count, dtype: int64
```





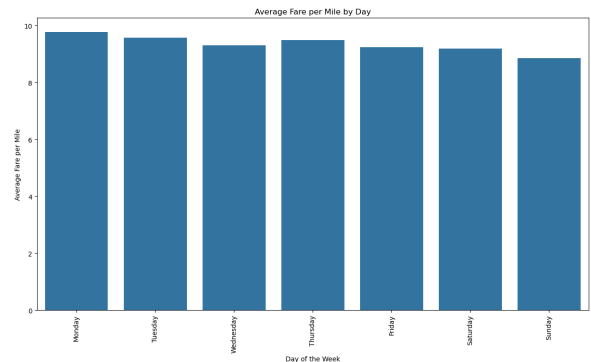
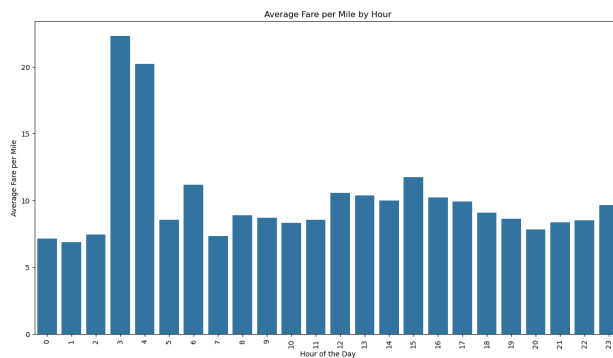
- Hourly pickup and dropoff trends for top 10 zones

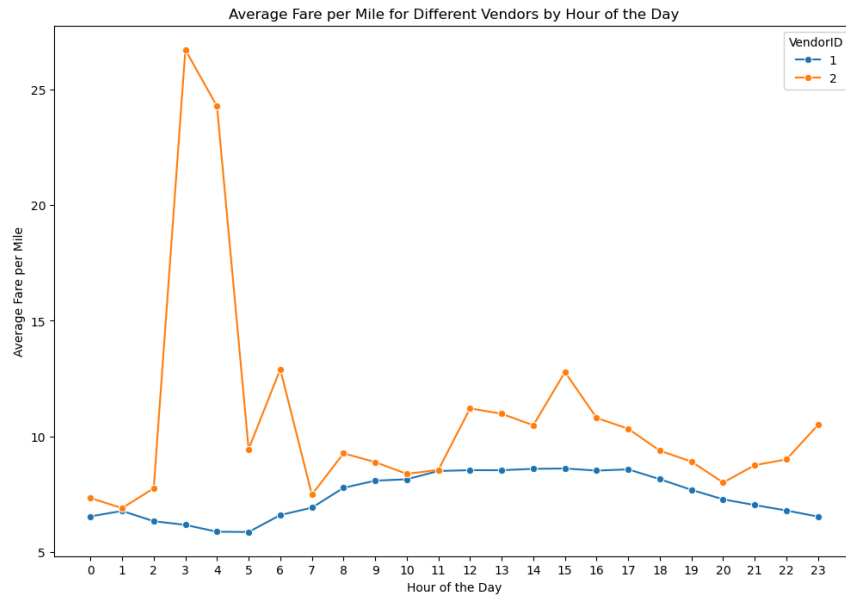




Pricing Strategy

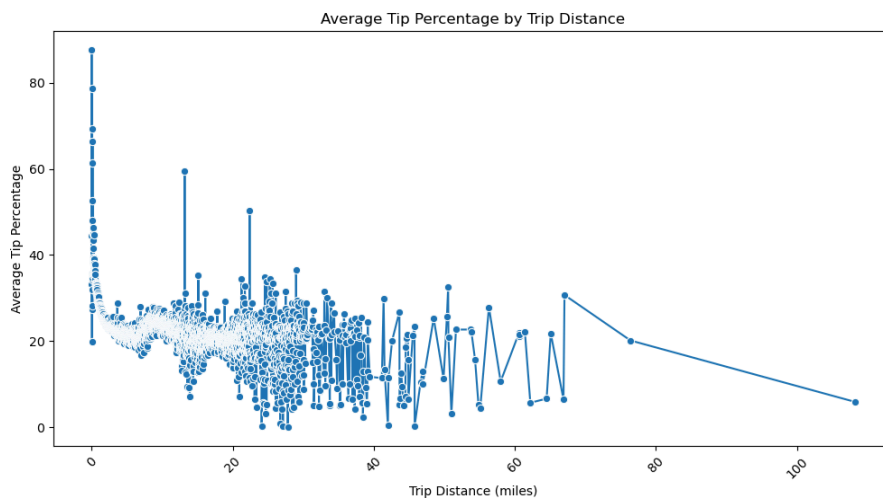
- The analysis shows that fare per mile changes across different hours of the day. The highest values are seen during early morning hours, especially around 3 AM and 4 AM, possibly due to lower trip counts, shorter rides, or nighttime surcharges.
- For most daytime and evening hours, the average fare per mile stays fairly stable between 8 and 12. Day-wise trends show only small differences, with Monday and Tuesday having slightly higher values, while Sunday has the lowest.
- Vendor 1 maintains more consistent fare-per-mile rates throughout the day, whereas Vendor 2 shows larger fluctuations, particularly during early morning hours. Overall Vendor 2 generally records slightly higher fare-per-mile values compared to Vendor 1.

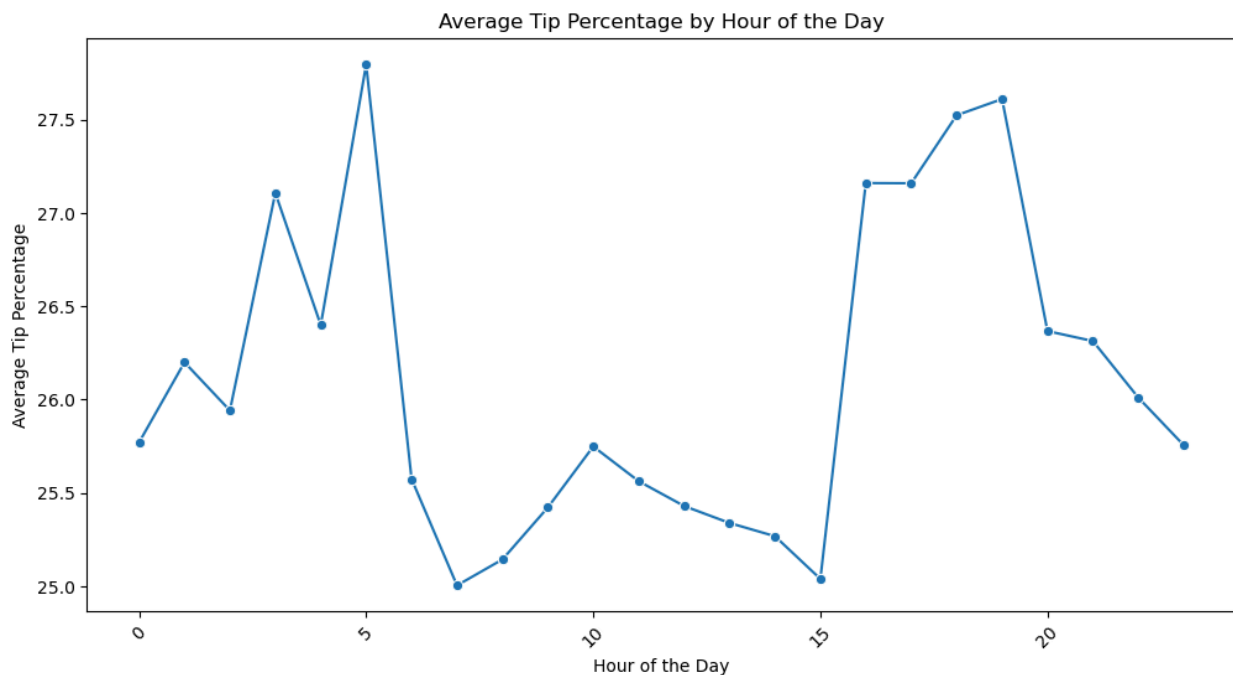
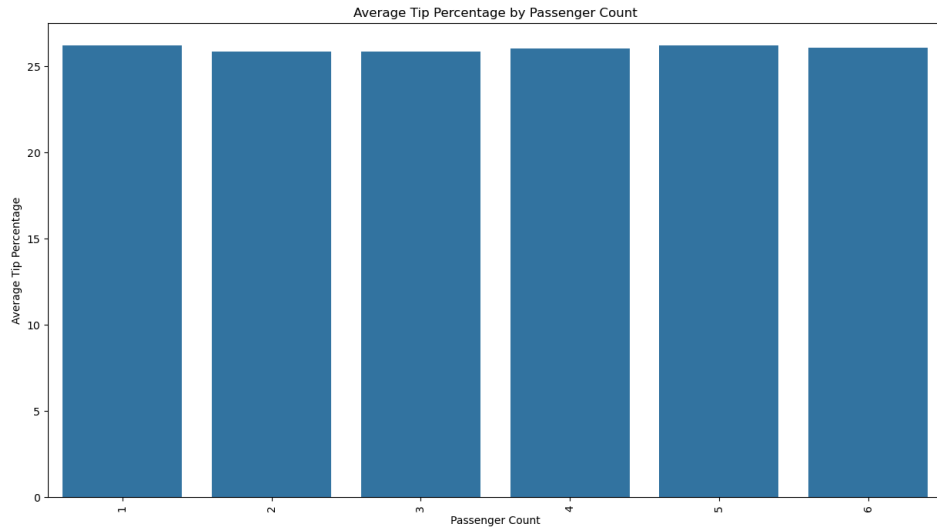




Customer Experience and Other Factors

- The charts show that tip percentage is influenced by trip distance, passenger count, and time of day.
- For trip distance, the average tip percentage generally decreases as distance increases, although there are fluctuations for longer trips due to fewer observations in those ranges.
- The passenger count analysis shows only small differences in average tip percentage across different group sizes, suggesting that passenger count has limited impact on tipping behaviour.
- The hourly trend indicates that tip percentages vary throughout the day, with slightly higher values observed during certain evening and early morning hours. This may be related to travel demand, trip purpose, or customer behaviour during those periods.

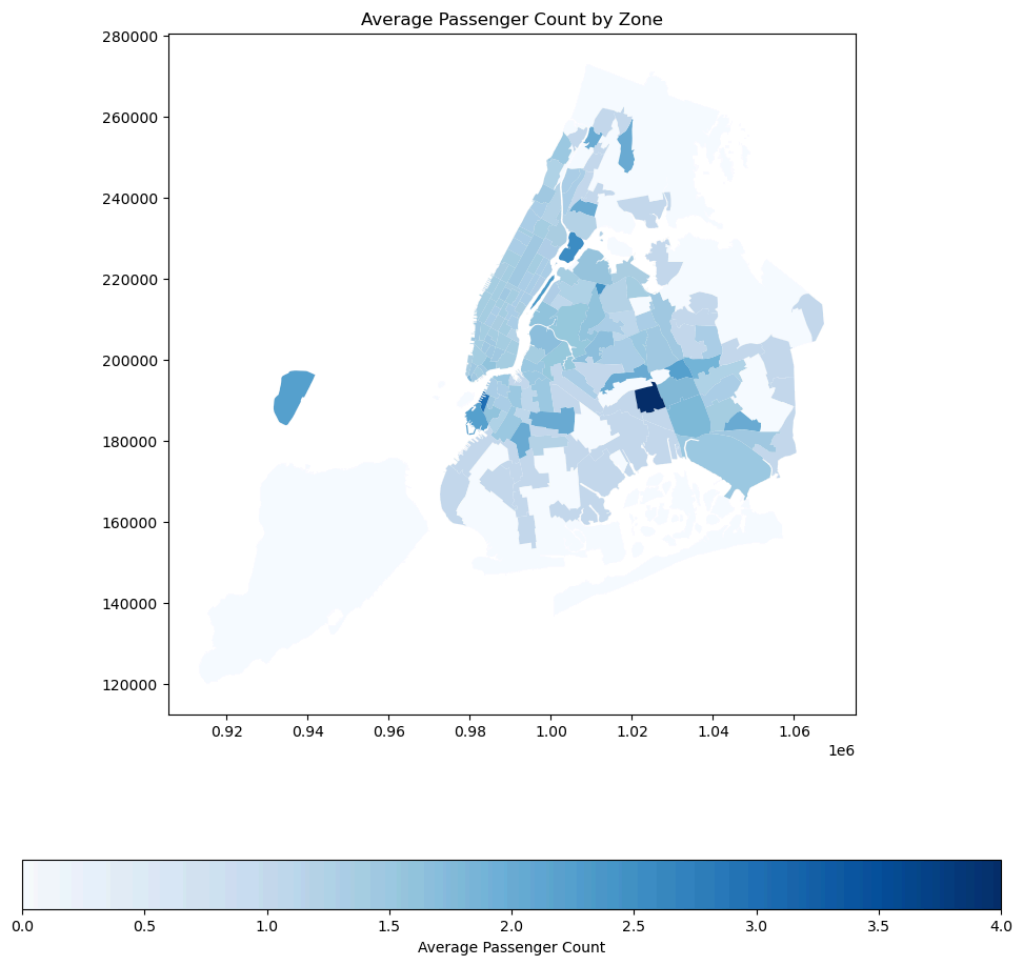
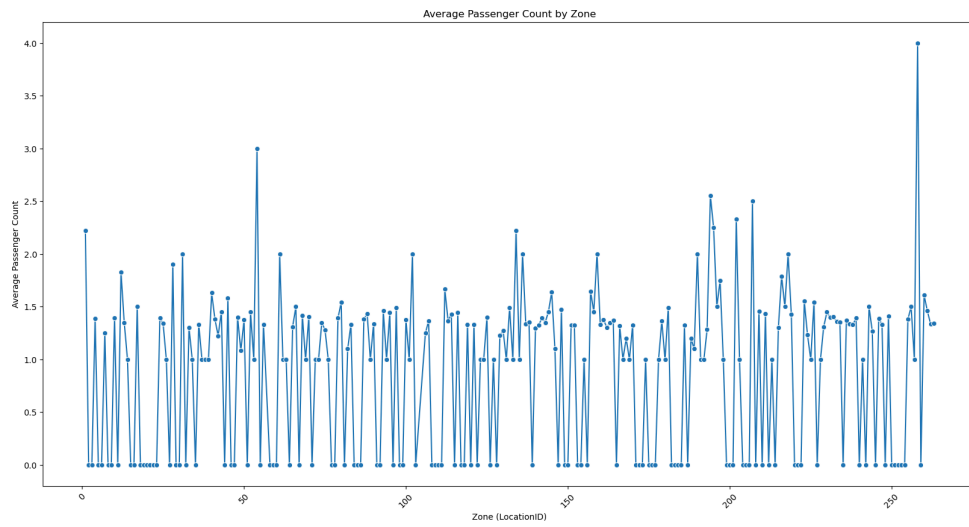




Average Passenger count by zone

- The analysis shows that average passenger counts vary across different taxi zones, although most zones have relatively low average passenger counts, generally between 1 and 2 passengers per trip.
- A few zones show noticeably higher averages, which may indicate areas with more group travel, airport traffic, or tourist activity. Several zones also record very low or near-zero averages, possibly due to lower trip volumes or limited activity in those locations.

- Overall, the variation across zones is moderate, suggesting that most taxi trips in NYC are typically taken by individual passengers or small groups.



Surcharges Analysis

- The chart shows that surcharge frequency varies significantly across different hours of the day. Surcharges are applied less frequently during early morning hours, especially between 3 AM and 5 AM, when trip activity is lower.
- The number of surcharges gradually increases during daytime and evening hours, with the highest frequency observed around 6 PM. This pattern aligns with peak travel periods when taxi demand and traffic congestion are generally higher.
- The analysis also shows that surcharge activity is more common in busy commercial and transportation zones such as Midtown Center, Upper East Side, JFK Airport, LaGuardia Airport, Times Square, and Penn Station areas. These locations experience high trip volumes and traffic flow throughout the day.

Frequency of each surcharge:

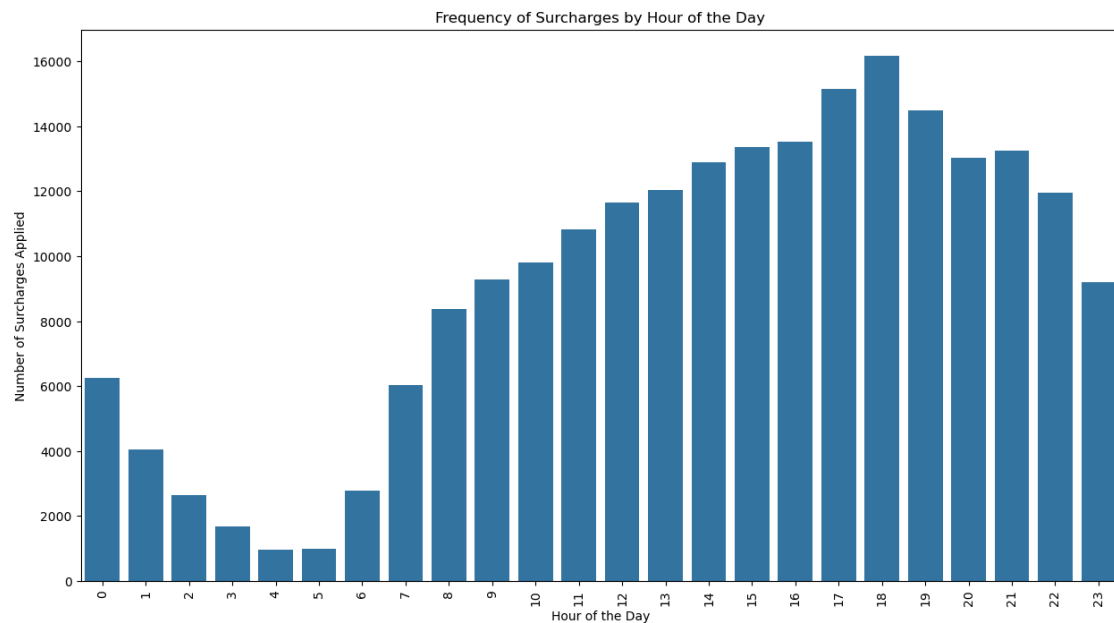
	Surcharge	Frequency
0	extra	138077
1	congestion_surcharge	209357
2	airport_fee_combined	18182
3	improvement_surcharge	220479

Top pickup zones with surcharges:

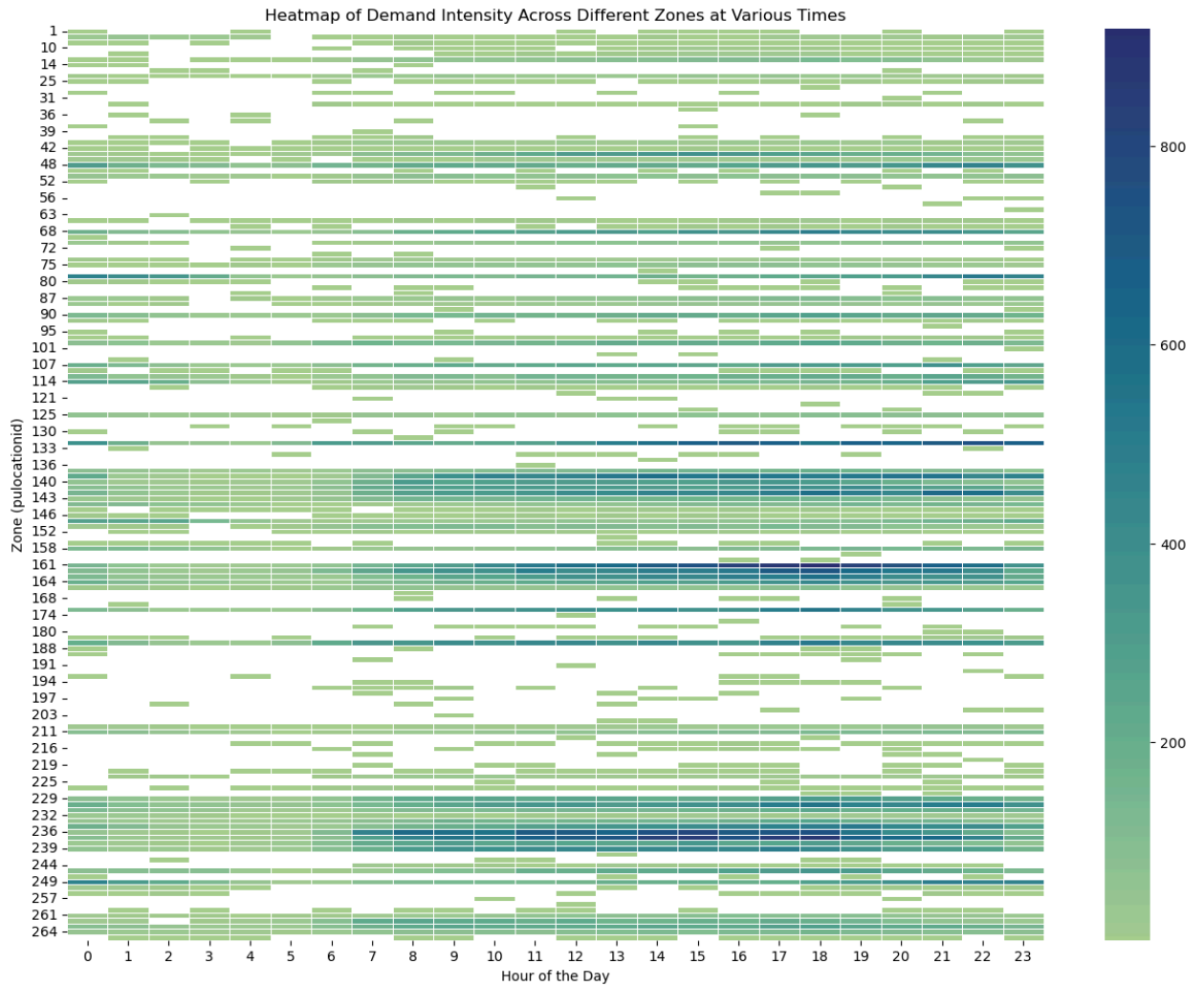
	pulocationid	surcharge_count	LocationID	zone
146	237	10949	237.0	Upper East Side South
97	161	10506	161.0	Midtown Center
75	132	9918	132.0	JFK Airport
145	236	9788	236.0	Upper East Side North
98	162	8260	162.0	Midtown East
81	138	8153	138.0	LaGuardia Airport
111	186	7541	186.0	Penn Station/Madison Sq West
84	142	7497	142.0	Lincoln Square East
140	230	6907	230.0	Times Sq/Theatre District
105	170	6613	170.0	Murray Hill

Top dropoff zones with surcharges:

	dolocationid	surcharge_count	LocationID	zone
229	236	10751	236.0	Upper East Side North
230	237	9687	237.0	Upper East Side South
155	161	8579	161.0	Midtown Center
164	170	6787	170.0	Murray Hill
232	239	6570	239.0	Upper West Side South
136	142	6408	142.0	Lincoln Square East
156	162	6273	162.0	Midtown East
135	141	6201	141.0	Lenox Hill West
223	230	5992	230.0	Times Sq/Theatre District
66	68	5616	68.0	East Chelsea



- Heatmap of Demand intensity across different time zones at various time



Conclusions

Final Insights and Recommendations

Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies:

- Allocate more cabs during peak travel hours, especially during morning and evening periods when trip demand is highest.
- Increase cab availability in high-demand pickup and dropoff zones to reduce passenger waiting time and improve trip allocation.
- Position more cabs in zones with consistent nighttime activity to reduce idle time and improve service availability.

- Monitor hourly demand patterns and dynamically redistribute cabs across zones based on real-time demand trends.
- Use historical demand trends and heatmap analysis to improve route planning and cab deployment efficiency.

Suggestions on strategically positioning cabs across different zones:

- The analysis identified several zones with consistently high trip activity throughout the day.
- High-demand commercial, airport, and central city zones should maintain higher cab availability during busy hours.
- Cab positioning should be adjusted dynamically based on hourly and daily demand fluctuations.
- Maintaining a balance between short-distance and long-distance trip zones can help improve driver utilization and reduce empty travel.

Recommendations for pricing strategy improvements:

- Monthly revenue trends show comparatively lower revenue during July, August, and September. Promotional offers or discounts during these months may help improve demand.
- Fare per mile varies across different hours of the day, suggesting that time-based pricing strategies may help optimize revenue during peak periods.
- Premium pricing can be applied in high-demand zones and peak traffic hours while maintaining competitive pricing during low-demand periods.
- Trip duration, traffic conditions, and waiting time appear to influence fare values and can be considered for future pricing optimization.
- Predictive analytics and demand forecasting models can help support more effective dynamic pricing strategies in the future.